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Research article

Influence of laser beam intensity maxima and their relative positions on the keyhole during multi-beam laser deep penetration welding

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ABSTRACT

The intensity distribution of the laser beam during deep penetration laser welding strongly affects the process behavior. While the influences of several specific distributions have been studied, the almost infinite number of possible beam profiles makes it impractical to examine each experimentally. Therefore, a more profound understanding of how intensity maxima at defined keyhole locations interact with the keyhole is required to enable an a priori design of task-specific intensity distributions in the future. This study investigates the hypothesis that the characteristics of individual laser beam processes can coexist independently within a combined process, provided that the additional intensity maximum interacts only locally at the keyhole bottom. An optical system was used that coaxially combines a multi-mode laser beam and a single-mode laser beam, allowing the intensity of the single-mode laser beam to be positioned within the larger keyhole generated by the superimposed multi-mode laser beam in mild steel. The effects on keyhole morphology, weld formation, and process stability were analyzed using optical coherence tomography data, high-speed imaging recordings, and metallographic cross-sectional micrographs. The results confirm the hypothesis by demonstrating that an additional intensity maximum at the keyhole bottom leads to the formation of a secondary keyhole tip while leaving the primary keyhole shape largely unaffected – creating a stable "keyhole-in-keyhole" process. In contrast, placing the additional intensity on the primary keyhole's front wall alters the inclination angle, shifts the primary keyhole bottom, and increases the occurrence of process irregularities such as spatter formation. These findings advance the understanding-based portfolio for the targeted manipulation of keyholes and represent a step toward an a priori design of laser beam intensity distributions.

1. Introduction

Deep penetration laser beam welding is a key technology in modern manufacturing that offers various advantages, such as low heat input into the workpiece caused by the achievable high aspect ratios of the weld seam cross sections as well as high welding speeds compared to other welding processes. The welding process can be influenced by parameters such as the power of the laser beam, its wavelength, and the welding speed, with the most important parameters probably being the maximum intensity of the laser beam and its distribution.

Over time, different approaches have been used to adjust different static intensity distributions: the superposition of multiple individual beams, spatial beam shaping by using optical elements, the use of multi-core fiber laser sources, and the use of coherent beam combining. The relevance of these approaches does not primarily lie in the technologies themselves but in the fact that they enable a wide range of fundamentally different energy field distributions within the interaction zone. Numerous published studies describe investigations on the influence of specific changes in intensity distribution on the process behavior. The concepts and the exemplary results achieved are briefly presented below in the subchapters, which are structured according to the method.

In the late 1970s, it was demonstrated that combining two non-coaxially and slightly offset arranged electron beams could increase the achievable welding speed without the previous occurrence of humping by changing the melt flow [1]. According to [2], it was later possible to demonstrate this positive influence on welding speed for

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welding with two CO_2 laser beams. A comparable combination of CO_2 laser beams with a common superposition, which could thus be adjusted in terms of the intensity distribution, was used to achieve higher total power [3]. This so-called multiple spot laser welding technology has been used repeatedly over the years, either in one common superimposed spot or with two or more spots that were slightly offset. This approach enabled process improvements such as a higher gap bridging capability due to an increased amount of molten material [4] and producing weld seams in high-carbon steels with higher tensile strength using a post-heating treatment [5]. By generating an elongated keyhole, which achieved continuous vapor flow and increased keyhole stability, higher weld seam quality was achieved [6]. The improved vapor flow due to the modified intensity distribution caused the reduction of process pores and blow holes in aluminum weld seams [7]. As a result of the superposition of two different laser beams, a keyhole stabilization and an attributed decrease in material loss were observed during welding [8]. A coaxial and concentric combination of two individual laser beams with comparable wavelengths was utilized for laser beam welding through the development of a specially designed optical system [9]. By combining a 1030 nm disk laser beam and a highly focused 1080 nm single-mode laser beam using this system, the so-called "keyhole-in-keyhole" formation in the form of a maximum in weld depth was observed in a previous study of the main authors [10].

A common method for setting almost any intensity profile with just one laser beam is to use beam shaping optical elements and lens groups that change its phase profile and intensity. This technology also enabled similar process improvements. It was demonstrated that the use of a shaped "Top hat" beam compared to a Gaussian beam resulted in different keyhole shapes and properties [11]. A diffractive optical element (DOE) was used to split one high power single mode fiber laser beam into three to seven separate beams arranged close to each other for producing a common rectangular deep penetration weld seam cross section using much less total power compared to heat conduction welding [12]. Similar experiments with a laser beam divided into four or nine separate beams showed, in addition to the aforementioned effect, an increased gap bridging ability, an improvement in process stability, and spatter formation, particularly with symmetrical intensity distributions [13]. A spatter reduction due to a stabilized keyhole opening was achieved for stainless steel by using a ring-shaped intensity profile [14].

Multi-core fiber laser systems offer intensity distributions consisting of a core and a surrounding ring intensity, between which the total power of the laser source can be flexibly distributed by controlling how much power is supplied to the core fiber and the concentrically surrounding ring fiber. Depending on the distribution of power, advantages for the process were identified, such as reduced spatter due to an enlarged keyhole opening [15], a widened melt pool [16], and the possible use of higher process speeds [16]. Investigations on affecting the formation of intermetallic compounds demonstrated a grain refinement due to lower peak temperatures and the altered migration of certain elements [17]. In summary, it was shown that the adapted core-ring intensity distributions can cause changes in the process regime and the process behaviors [18].

The most recently developed beam shaping technology for use in high-power laser beam applications is coherent beam combining (CBC). The active control of optically phased arrays of individual collimated high-power fundamental mode fiber lasers allows the generation of a combined beam with a common phase front and an almost completely customizable intensity distribution being variable in real time [19]. Adjustments of the entire optical phase with frequencies of up to 80 MHz enable the setting of quasi-static or dynamic beam profiles [19]. Using this technique, the weld seam quality in copper material was examined for 18 different beam profiles, from a single Gaussian laser beam intensity to variants like an "hourglass" profile [20]. Profiles such as 3 diagonal parallel lines were found that produced significantly higher weld seam qualities than a single Gaussian beam. The need to develop individual shapes for each material was concluded to be successful in process development [20]. Other beam profiles, such as a combination of two lines and a point, changed the weld shape from narrow-deep to wide-shallow in duplex steel, whereas the opposite trend was observed in austenitic steel [21], which also confirms the demand for task-specific intensity distributions. In addition, many further tests were carried out using dynamically varying beam profiles, the effects of which on the keyhole are even more complex.

The current state of research indicates that there is a growing number of technologies on the market for creating almost completely customizable intensity distributions and that some of them can have a positive effect on specific deep penetration laser welding processes. The reasons why certain intensity distributions stabilize the process and how exactly they influence the keyhole could only be clarified retrospectively for some specific distributions after conducting and analyzing corresponding experiments. What remains largely unaddressed is a systematic investigation of how the relative position of an intensity distribution in the keyhole affect its behavior and process stability. As a result, the physical mechanisms by which specific intensity distributions interact with different regions of the keyhole are not yet sufficiently understood. Given the almost infinite possibilities for designing the intensity distribution and the ever-widening range of applications and welding tasks, it is not economical to investigate the effects of numerous intensity distributions for each of these tasks individually. Rather, an *a priori* development of an intensity distribution designed for the specific welding task is necessary to minimize the effort involved in searching for suitable welding parameters by trial and error. To close this research gap and approach the possibility of an a priori design of the intensity distribution, extensive knowledge of the effect of individual characteristics of intensity distributions on the keyhole, and thus the process behavior, is necessary.

This study therefore aims to contribute to the comprehension-based, targeted local manipulation of the keyhole by systematically investigating the influence of a laser beam intensity maximum and its position within a laser beam induced keyhole. Rather than proposing or optimizing a specific beam profile for a predefined application, the focus is placed on identifying fundamental interaction mechanisms between localized energy input and the keyhole behavior. The work is based on the hypothesis that, if the additional intensity maximum is placed on the keyhole bottom, a second, significantly smaller keyhole volume can be produced and positioned largely independently of the main primary keyhole. This would allow for a targeted intensity distribution based on the results of the individual processes. It is simplistically assumed that two differently dimensioned keyhole shapes can be "added" to illustrate this concept. While irradiation at the keyhole bottom is expected to result in high local energy absorption due to favorable Fresnel conditions and thus a localized keyhole extension, irradiation of the keyhole front wall is assumed to couple more strongly to the primary process through reflection and energy redistribution within the keyhole.

To examine this hypothesis, the study investigates whether a coaxial secondary laser beam can generate a local keyhole extension at the bottom of a primary keyhole formed by a primary laser beam. If the hypothesis holds true, the additional keyhole section induced by the secondary laser beam could also be positioned off-center relative to the keyhole bottom, as it would not depend on the central intensity maximum of the primary laser beam or on multiple reflections within the primary keyhole.

This was investigated by using an optical system [9] that coaxially combines a multi-mode (MM) laser beam and a single-mode (SM) laser beam and can shift their foci in all three spatial directions relative to each other. In contrast to coherent beam combining, which relies on active phase control to generate a coherent beam with a predefined global intensity distribution and often undesirable secondary intensity maxima, the present approach superimposes two mutually incoherent beams with independently adjustable power and focal position to introduce a localized additional intensity maximum within an existing keyhole. This allowed the MM laser beam to create a large keyhole in which an intensity maximum of the SM laser beam was set and shifted inside the keyhole throughout the experiments. The resulting effects on the keyhole shape and its stability were investigated using optical coherence tomography, high-speed imaging, and metallographic analyses.

2. Experimental

The multi-laser-optics (MLO) shown in Fig. 1 and described in [9] in more detail superimposed a MM laser beam, a SM laser beam,min an and a beam from an optical coherence tomography system (OCT) coaxially. A primary keyhole was generated by the 1030 nm MM disk laser beam (TruDisk 12002 by Trumpf) with a resulting beam diameter on the sample surface of 1586 µm, a power of 4000 W up to 5000 W, a constant welding speed of 2 m/min, and a welding length of 50 mm. The coaxial SM fiber laser beam (TruFiber 2000P from Trumpf) with a wavelength of 1080 nm and a focus diameter of 42 µm switched on by means of a distance-based sensor after the MM laser beam had already created a 5 mm weld seam and formed the primary keyhole. This resulted in a length of 45 mm, or about 1.35 s, in which the SM laser beam interacted inside the primary keyhole. The specifications, caustics, and intensity distributions of the laser beams are displayed in Table 1. The MLO was protected from the vapor plume and spatter by a cross-jet at a height of 53.5 mm above the sample surface. The sample material was steel S235JRC. The chemical composition is given in Table 2. The sample dimensions were 60 × 40 x 10 mm^3 . On a single specimen, each configuration was repeated 5 times over the sample width of 40 mm. After each individual welding operation, the specimen was cooled to approximately room temperature using the water-based cooling system of the copper clamping fixture. The water temperature was 22°C with a flow rate of approximately 6 l/min. The cooling time was about 5 min.

The processes and analyses shown in the flowchart in Fig. 2 were carried out to investigate the influence of laser beam intensity maxima and their positions in laser welding induced keyholes. These are described in detail below.

Metallographic cross-sections were taken with a distance of 25 mm to the sample edge (see Fig. 3) and etched with 1% nitric acid. Depending on the process parameters, the seam cross-sections were divided into a primary part, which was mainly generated by the MM laser beam, and a secondary part, which was mainly generated by the SM laser beam and clearly differs from the shape of the primary part (see Fig. 3). This was then used to determine the measured values of the primary and secondary weld depth, width, and seam cross-section area. To determine the eccentricity of the secondary seam cross-section area first, the geometric center of the weld seam was defined by halving the width of the weld seam and transferring this position from a defined weld edge into the cross-sectional image. This position was used as the reference for the weld center. Subsequently, the horizontal distance between the weld center and the visually identifiable center of the secondary melt pool cross-section area was measured. This distance corresponds to the eccentricity ΔE (see Fig. 11).

It can be evaluated whether the individual processes can simply be added together in the combination by subtracting the theoretical sum of the values of the individual processes from the actually measured total values of the combined process (see Fig. 4).

Two synchronized high-speed cameras were used for process monitoring. High-speed camera No. 1, the i-Speed7 (iX Cameras Ltd.) with an AF 180 mm 3.5 Di LD Macro 1:1 SP lens (TAMRON Europe GmbH) and a connected BN810 narrow near-IR bandpass filter, observed the keyhole opening area. It was positioned opposite to the welding direction and in a 35° tilt angle so that an image section of about 3.5 × 4.2 mm^2 around the keyhole opening area could be captured with a resolution of 512 × 512 pixels and a varying frame rate of 20 kHz to 50 kHz. Since this alignment only allowed a view of the bottom and rear walls of the keyholes (see Fig. 1), some experiments were repeated in which the feed direction and thus the welding direction were reversed, allowing the camera to capture the front wall of the keyhole. The keyhole opening area was tracked using a MATLAB (MATrix LABoratory by MathWorks)-based algorithm that outputs the size of the keyhole area A_{Kh}, the keyhole width w_{Kh} and the keyhole length l_{Kh} per frame, which was set by means of a brightness threshold (see Fig. 5b and c). The mean values ($\overline{A}_{Kh}$, $\overline{w}_{Kh}$, and $\overline{l}_{Kh}$) and the standard deviations over a period of one second were then determined from the values per individual frame. The analyzed frames captured over a period of one second (Fig. 5d) were combined into an averaged image (Fig. 5e) to visualize this average keyhole opening area.

The vapor plume and spatter were recorded using the second high-speed camera FASTCAM NOVA S12 (Photron), with an AF 180 mm 3.5 Di LD Macro 1:1 SP lens (TAMRON Europe GmbH) and a connected BN810 narrow near-IR bandpass filter. This camera No. 2 recorded at a frame rate of 20 kHz and a resolution of 512 × 512 pixels. It was positioned perpendicularly to the welding direction and at a 90° tilt angle to capture an image section of about 62.75 × 62.75 mm^2 around the

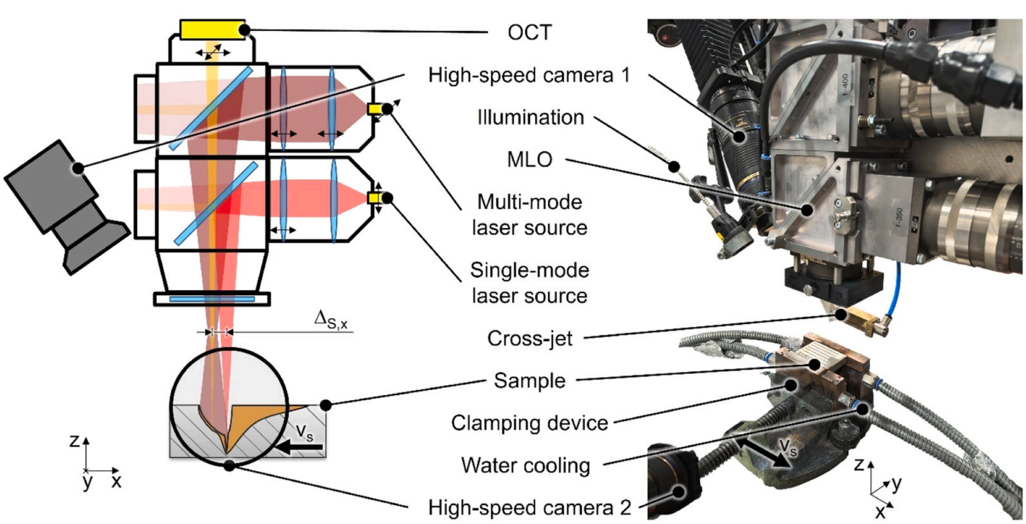


Fig. 1. Experimental setup with multi-laser optics (MLO) for the coaxial combination of a MM laser beam, a SM laser beam, and the beam from an optical coherence tomography system (OCT) with the option of shifting the beams and their focal positions in three dimensions relative to each other; v_s is the welding speed and direction, which could be changed to allow the high-speed camera No. 1 to observe the keyhole bottom or the keyhole front wall.

Table 1
Specifications of the laser beams used.

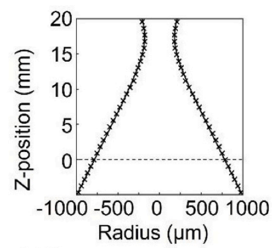
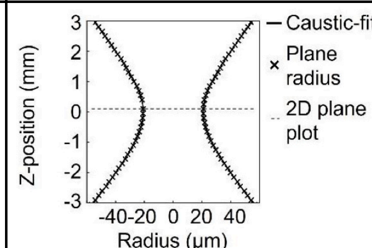
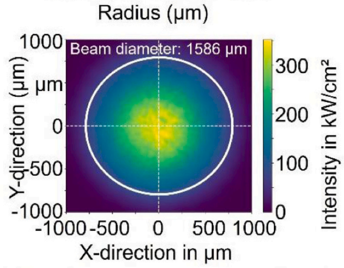
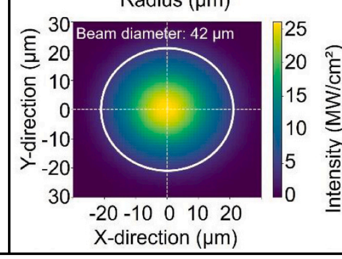
Laser source	Multi-mode laser source TruDisk12002	Single-mode laser source TruFiber 2000P
Wavelength	1030 nm	1080 nm
Beam quality M^2	23.31	1.05
Beam parameter product	7.89 mm•mrad	0.36 mm•mrad
Rayleigh length	4.05 mm	1.22 mm
Angle of incidence	0°	0°
Focus diameter	358 μm	42 μm
Intensity distribution	Gaussian	Gaussian
Focal plane	+17.5 mm	see Table 3
Beam diameter	on sample surface: 1586 μm	on primary keyhole bottom: 42 μm
Caustic		
Intensity distribution (2D plane plot at z = 0 mm)		

Table 2
Chemical composition (wt%) of S235 according to [22].

Material	Chemical composition (wt%)								
	Fe	Al	Ni	Mn	Cu	Si	Cr	C	Ti
Steel S235JRC	bal.	0.002	0.130	0.520	0.230	0.200	0.160	0.080	0.001

process zone (see Fig. 6a). Spatter tracking was performed using a MATLAB-based algorithm, which, after defining the region of interest and excluding the area of the vapor plume (Fig. 6b), creates a binary image using a gray value filter, from which the spatters are identified and tracked using the white pixels (Fig. 6c). It outputs the mean size of each individual spatter projection in mm^2, which, assuming that the spatter are spherical, can be used to determine the spatter volumes. From the mean number of spatter per second multiplied by the mean spatter volume, the mean spatter amount in mm^3 per second was determined, which takes both parameters into account. The smallest detectable spatter diameter was 0.12 mm.

The illumination laser source CAVILUX HF (Cavitar Ltd.) was only used in experiments capturing the keyhole opening area and was shut off when recording the vapor plume and spatter.

Averaged keyhole shapes were measured using OCT to determine the point of impact and the effect of the SM laser beam inside the keyhole. The OCT system was a Laser Depth Dynamics LD-600 (Laser Depth Dynamics Inc., now part of IPG Photonics Corporation). It uses a wavelength range of 800 nm to 900 nm, a focus diameter of approximately 100 μm and has a measuring frequency of up to 250 kHz. With its ability to move the measuring beam in the plane with a maximum measuring range of 5 × 5 mm^2 using a 2D scanner, it was used to scan the keyhole as well as parts of the melt pool and the sample surface and thus obtain 3D data. The position of the OCT beam was adjusted so that its reference point was in the center of the intensity of the MM laser beam ($\Delta_{S,x} = \Delta_{S,y} = 0$) and the focus was set on the sample surface. The scan time was 1.35 s in each case and was initiated simultaneously with the power output of the additional SM laser source. The maximum possible frequency during the scanning process was 111 kHz, so 150,000 measurement points were recorded for each parameter during this time. In one repetition of the process parameter, an area of 3000 × 3000 μm^2 was scanned. In doing so, the sample surface, the keyhole, and the keyhole bottom become visible through a few measurement points at greater depths (see Fig. 7a). In case of another repetition of the respective process parameter, the measuring range was reduced slightly. 3000 μm were scanned in the x-direction and 500 μm in the y-direction around the reference point, whereby more measuring points could be determined from the area of the keyhole bottom, and thus its relative position to the center of the MM laser beam could be determined more clearly (see Fig. 7b). For the remaining repetitions, smaller symmetrical areas around the keyhole bottom were then scanned to obtain even more data points from this area (Fig. 7c). By adding up all measured data

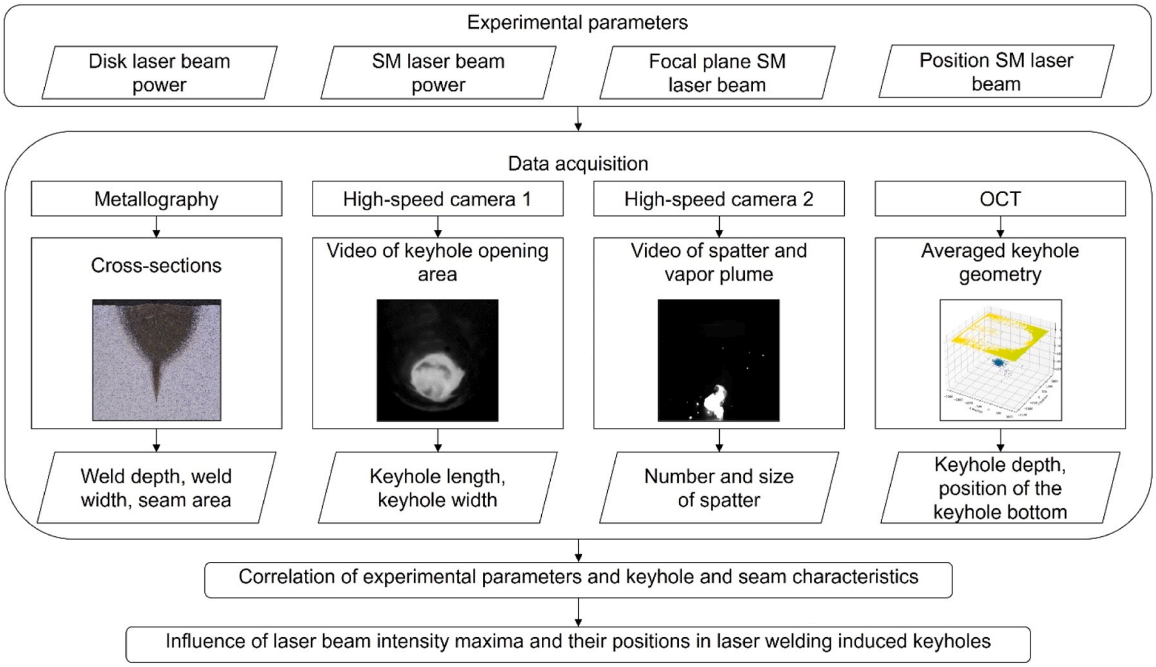


Fig. 2. Flowchart of the processes and analyses carried out to determine the influence of laser beam intensity maxima and their positions within keyholes induced during laser welding.

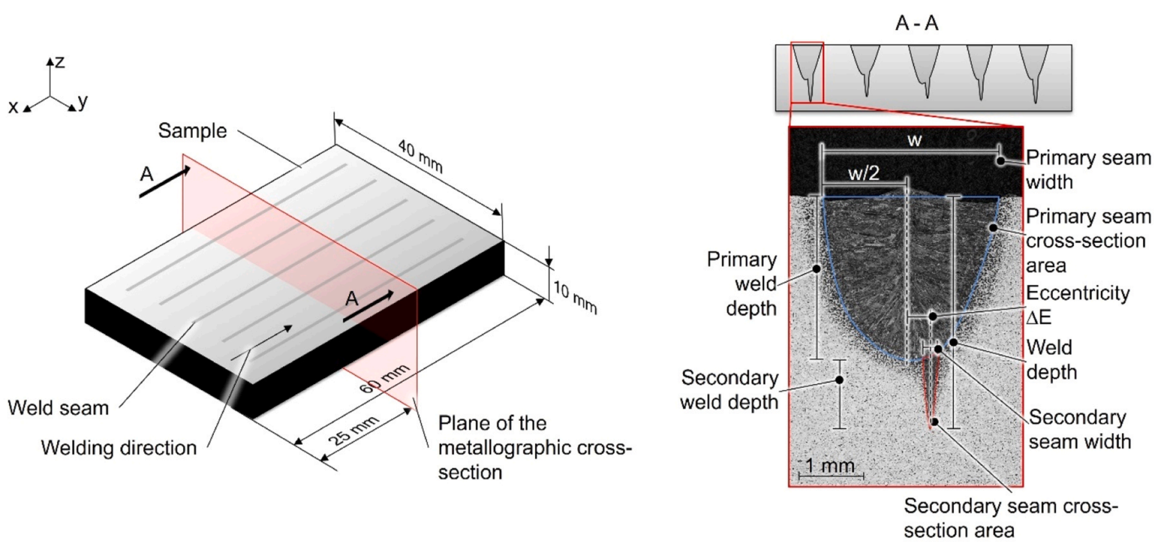


Fig. 3. Dimensions of the sample and measured values. The seam cross-section areas were divided into a primary part (mainly generated by the MM laser beam) and a secondary part (mainly generated by the SM laser beam).

points in one file, an averaged keyhole shape could be captured (Fig. 7d and e). A lower threshold of the backscattered light intensity level (signal quality) was used for the evaluation, which was determined in preliminary tests based on a stability criterion of the extracted keyhole shape (see supplementary material 01). The threshold was systematically varied between 0 dB and 18 dB, and both the mean distance of the primary keyhole bottom from the center of the primary laser beam and the mean keyhole depth were evaluated. Up to a threshold of 10 dB, no significant changes in either quantity were observed, whereas a pronounced reduction of noise points occurred from approximately 11 dB onward, which is also visible in the two-dimensional representations of the keyhole shape. For thresholds of 15 dB and higher, both quantities remained constant, and therefore 15 dB was selected as the lower signal quality threshold.

Based on the measured averaged keyhole shapes, further analyses could be carried out regarding the position of the keyhole bottom relative to the center of the MM laser beam. For the keyhole depth, a plot was created that shows all measured points in the Y-Z plane (see Fig. 8b). This data was then transferred to an average shifted histogram showing the number of measurement points over the depth with an interval width of 20 µm (see Fig. 8c). The maximum, which is located at a higher depth than that of the sample surface (at a depth of 0 mm), was defined as the average depth of the keyhole bottom. The thickness of the molten film at the bottom of the primary keyhole was determined based on the difference between the maximum weld penetration depth determined by metallography and the keyhole depth determined by OCT (see also the supplementary material 01 for a schematic representation of the calculation).

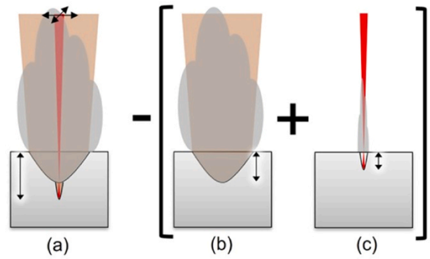


Fig. 4. Illustration of the calculation of a comparative value. The total weld depths and the seam cross-section areas (a) were compared to the theoretical sum of the results of the individual processes with only disk laser beam power (b) and only SM power (c).

For the position of the keyhole bottom relative to the center of the MM laser beam in x- and y-direction, only measured values that were determined using a symmetrical and consistent scan pattern were considered. In addition, the measured values were restricted so that only data points from the areas marked in red in Fig. 8d and g were used in the respective 2D x-z (Fig. 8e) and y-z plots (Fig. 8h), whereby data points that correlate with the sample surface are ignored. Using the data from these plots, average shifted histograms with an interval width of 20 μm were determined using only data measured below a depth z of 300 μm, ensuring that only points measured inside the keyhole were considered. The maximum to be found in the histograms was defined as the average relative distance of the keyhole bottom in X (Fig. 8f) and Y (Fig. 8i) to the center of the MM laser beam. Provided that there is a clearly identifiable peak, the standard deviation describes the statistical dispersion of the peak positions.

It should be noted that the complex coating of the beam-combining mirrors in the MLO leads to artifacts with regard to the OCT signal, which are visible in the form of so-called "echo surfaces" at welding depths greater than 3 mm. For this reason, only measuring points up to a depth of 3 mm were considered for OCT-based evaluations. This limitation affects only a small subset of the data used in this study. In particular, it is relevant only for the evaluation of the primary keyhole bottom position (Fig. 14), where two data points at 5000 W MM power exceeded the measurement range and were excluded. All other OCT-based data remain within the valid range.

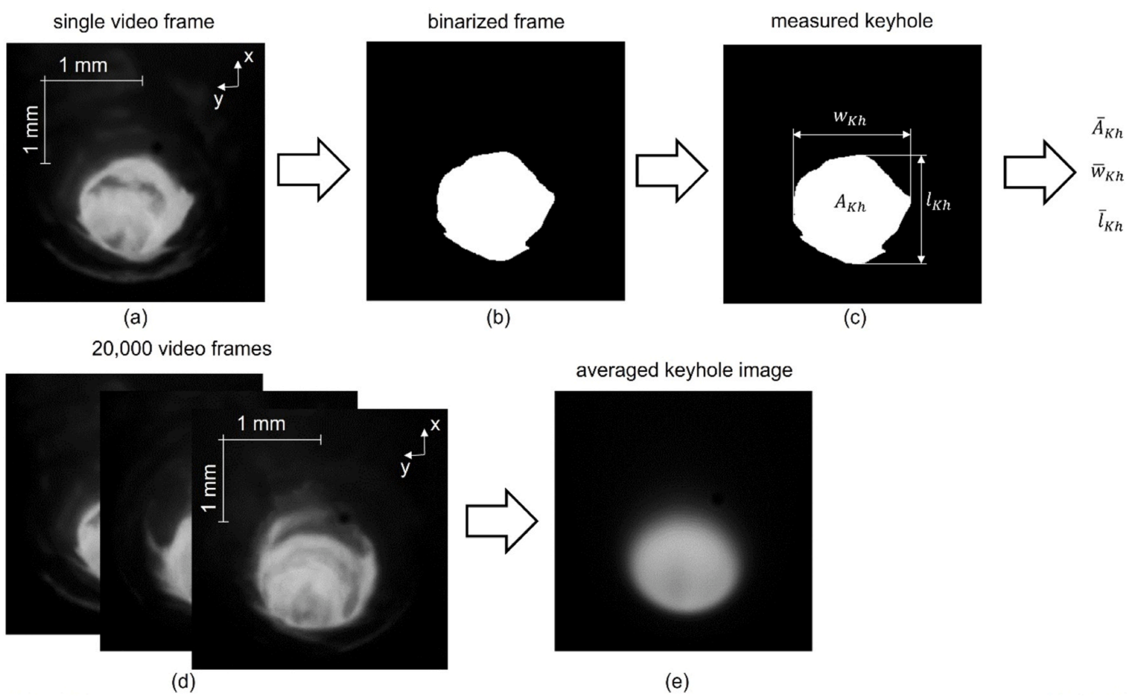


Fig. 5. Determination of the keyhole opening area values. (a) Original frame from a high-speed video used for keyhole area measurements. (b) Detected keyhole area. (c) Measured keyhole area A_{Kh}, keyhole width w_{Kh} and keyhole length l_{Kh} per frame, from which the mean values ($\overline{A}_{Kh}$, $\overline{w}_{Kh}$, and $\bar{l}_{Kh}$) are calculated over 1 s of welding time. (d) 20,000 images taken over a period of one second were combined to an averaged image (e) of the keyhole opening area.

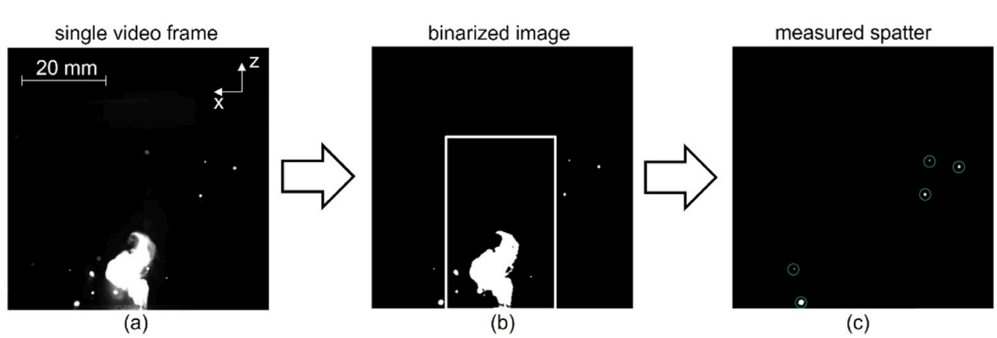


Fig. 6. Spatter detection. (a) Original frame from a high-speed video used for spatter tracking and visualization of the vapor plume. (b) binarized frame and selection of the region of interest for the spatter tracking algorithm with the exception of the vapor plume area. (c) resulting frame and tracked spatter.

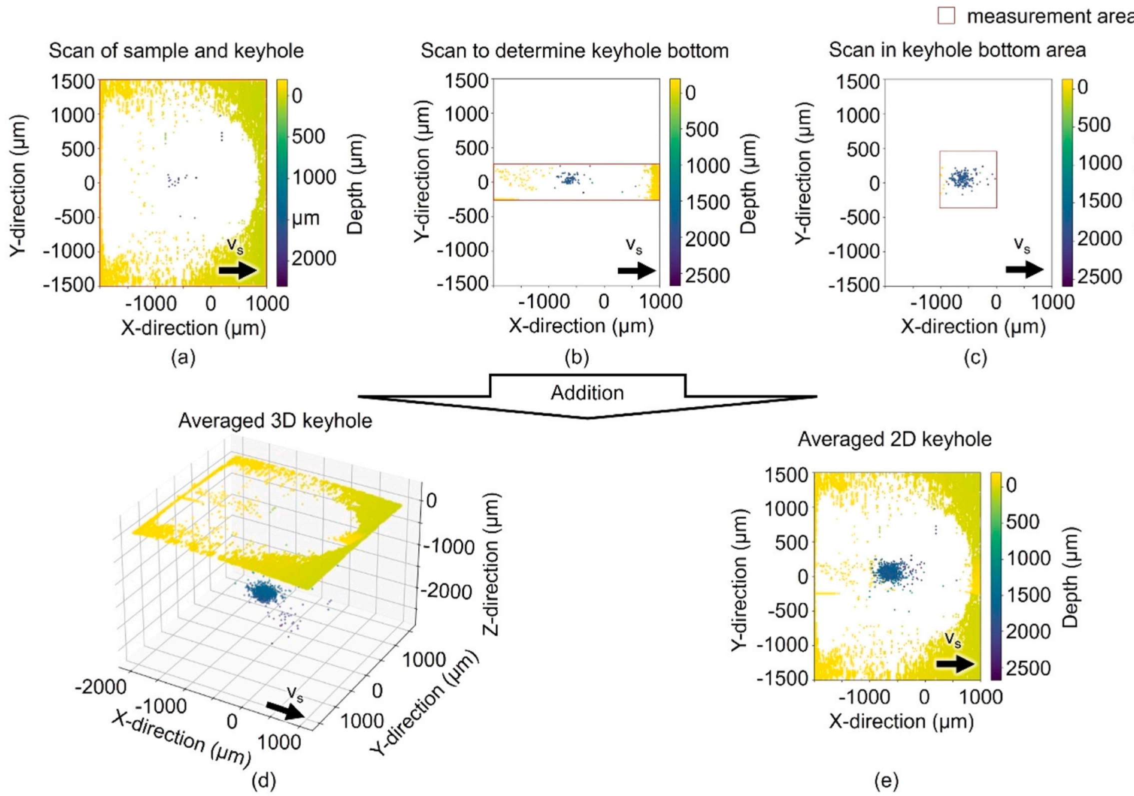


Fig. 7. Generation of an averaged keyhole shape image (d, e) by combining individual measurements (a, b, and c) using a scanning method of the OCT. In order to create one image of an averaged keyhole shape, 5 individual measurements were combined each time, which were recorded in different regions of interest (red rectangles) such as the entire keyhole and the surrounding sample (a), the area of the keyhole and its front and rear extensions (b) as well as the keyhole bottom (c).

The experimental plan is shown in Table 3. The MM laser beam power of 4000 W and the focal plane of + 17.5 mm were selected because this configuration produced a large and sufficiently deep primary keyhole, enabling reliable OCT measurements and high-speed imaging. An additional MM laser beam power of 5000 W was investigated to assess the influence of increased primary laser power. At this power level, the primary keyhole bottom could still be detected by OCT, while greater depths could no longer be reliably measured due to the previously described echo surfaces. During the variation of the MM laser beam power, the position of the SM laser beam was kept constant and aligned coaxially and concentrically with the MM laser beam (ΔS, x = ΔS,y = 0 µm). This configuration corresponds to the center of the MM intensity distribution and resembles a commonly used concentric core–ring intensity profile. In addition, this position is located at the front wall of the primary keyhole and was therefore selected as a reference case to enable a direct comparison with configurations in which the SM laser beam interacts with the keyhole bottom. In a second set of experiments, the MM laser beam power was fixed at 4000 W, while the position of the SM laser beam was systematically varied in the x- and y-directions. The x-offset $\Delta_{S,x} = -700$ µm was chosen based on OCT measurements of the primary keyhole shape, which indicated an average x-position of the primary keyhole bottom at approximately −690 µm (see Fig. 8f). The offset was not adjusted exactly to −690 µm, as a manual alignment accuracy better than ±10 µm would have required disproportionate effort and was considered unnecessary. This approximation is justified because the lateral extent of the primary keyhole bottom in x-direction is approximately 500 µm (see Fig. 8e and f), making deviations of this magnitude negligible with respect to the interaction region. The y-direction offsets were selected to probe different characteristic regions of the keyhole shape. At $\Delta_{S,x} = 0$ µm, a y-offset of $\Delta_{S,y} = -400$ µm was chosen, as high-speed imaging confirmed that this position is still located within the front wall of the keyhole. This ensured a pronounced interaction with the keyhole front wall, which would be less pronounced at smaller offsets. At $\Delta_{S,x} = -700$ µm, a y-offset of $\Delta_{S,y} = -250$ µm was selected because the histogram shown in Fig. 8i indicates that, at this distance from the keyhole center, isolated measurement points of the primary keyhole bottom are still present. This position therefore represents a laterally offset interaction with the primary keyhole bottom, maximizing off-center effects while minimizing interaction with the keyhole walls. Depending on the power of the MM laser beam and the resulting depth of the primary keyhole, the focal plane of the SM laser beam with a diameter of 42 µm was set accordingly. At a MM laser beam power of 0 W, the focal plane of the SM laser beam was set on the sample surface (0 mm) for reference experiments. For the cases in which the SM laser beam interacts with the bottom of the primary keyhole ($\Delta_{S,x} = -700$ µm), the depth of the primary keyhole directly corresponds to the interaction depth of the SM beam. Due to the inclination of the keyhole front wall and strong fluctuations, no reliable OCT data points are available for the interaction depth of SM beam positions on the keyhole front wall (cf. Fig. 8a). Therefore, at 4000 W and 5000 W MM laser beam power, the focal plane of the SM laser beam was set at the depth of the primary keyhole at −1.8 mm and −2.5 mm. In all experiments, the power of the SM laser beam varied from 0 W to 500 W to determine a possible power-dependent influence. The power was determined in preliminary tests, which indicated that fundamental effects occur within this power range. All process parameters were kept constant during each individual welding experiment. The laser sources were operated in continuous-wave mode. The welding speed was controlled by a motorized linear axis and remained constant during processing. This was ensured by providing an acceleration and deceleration distance of 100 mm before and after the specimen. Due to the short welding time of 1.5 s, a focus shift during welding could be excluded, which was further confirmed by dedicated focus shift measurements. The relative position of the SM laser

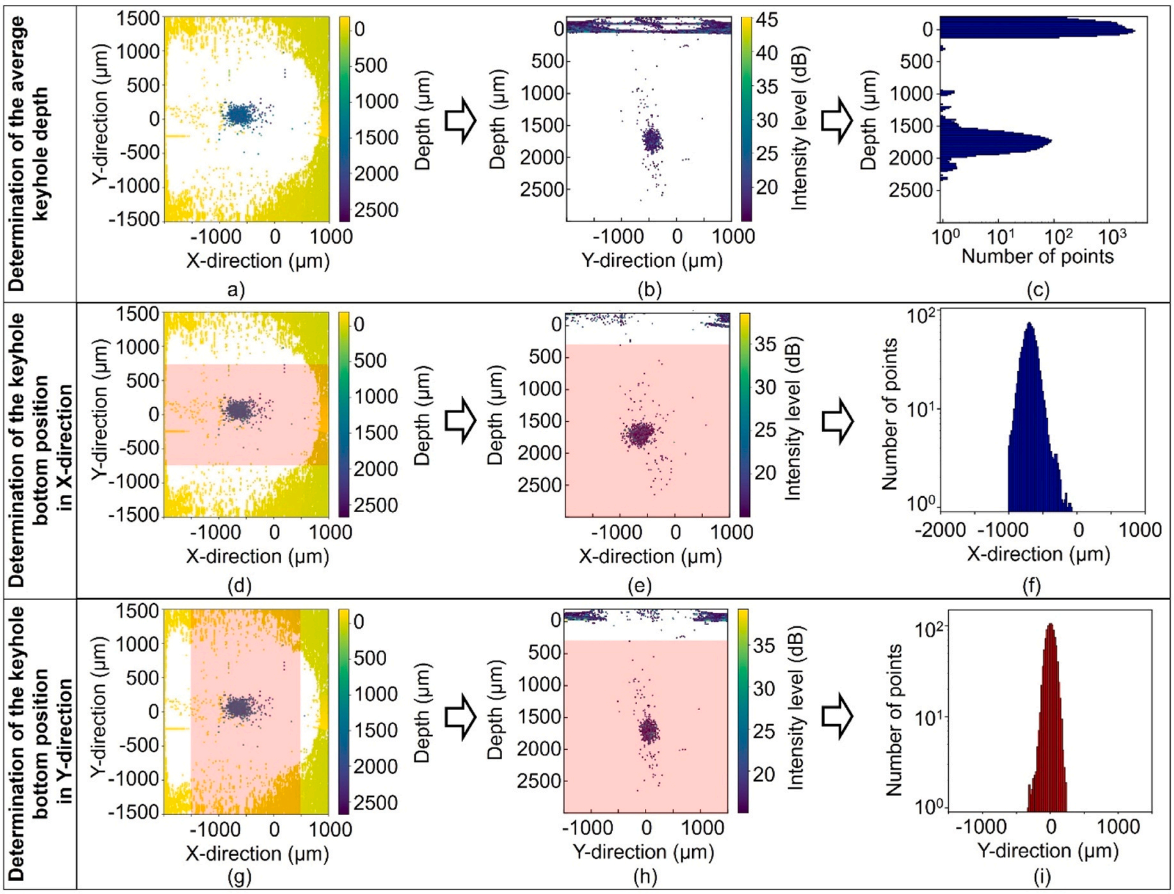
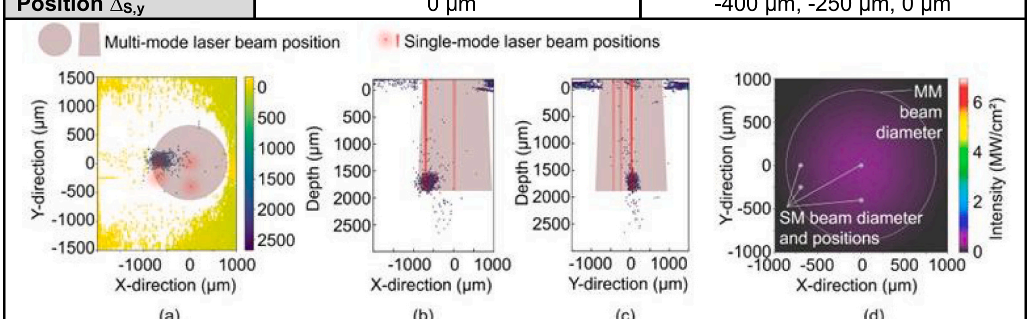


Fig. 8. Determination of keyhole features. Determination of the average keyhole depth (c) and the distance of the keyhole bottom to the center of the MM laser beam in x-direction (e) and y-direction (g) using the averaged keyhole shape (a) and the corresponding results for the primary keyhole at 4000 W power of the MM laser beam showing an average keyhole depth of 1790 μm, an average distance in x from the keyhole bottom to the MM laser beam center of −690 μm. Only data points from the areas marked in red were used in the histograms (c, f, i).

Table 3

Experimental plan. (a) – (c) True-to-scale representation of the superposition of beam positions, beam diameters, and caustics with the determined keyhole shape at a MM power of 4000 W and without additional SM power. d) Exemplary measured and mathematically superimposed intensity distributions with 4000 W MM power and 50 W SM power (the other used intensity distributions are provided in the supplementary material 01).

Laser source	Multi-mode laser source TruDisk12002	Single-mode laser source TruFiber 2000P
Focal plane	+17.5 mm	0 mm, -1.8 mm, -2.5 mm
Power	0 W, 4000 W, 5000 W	0 W, 100 W … 500 W
Position $\Delta_{S,x}$	0 μm	-700 μm, 0 μm
Position $\Delta_{S,y}$	0 μm	-400 μm, -250 μm, 0 μm



beam with respect to the MM laser beam was adjusted using beam diagnostics and verified after each weld before changing to the next parameter set. No measurable deviation was detected. Consequently, all relevant process parameters can be considered temporally stable for the duration of each welding test.

Changing the position of the SM laser beam (with constant power) within the larger intensity distribution of the MM laser beam, results in different intensity values being achieved within the focal diameter of the SM laser beam due to the superposition of the two beams. It was assessed whether the spatial superposition of the two laser beams influences the process to ensure that the experimental results can be compared reliably. The measured intensity distributions of both beam sources were mathematically superimposed by adding their intensity profiles while taking the respective focal positions into account. For this purpose, the superposition of the MM and SM laser beams was treated as an incoherent intensity superposition. Since the two laser sources are not phase-coupled, the local total intensity was calculated as the linear sum of the individually measured intensity distributions:

$$I_{total}(x,y)=I_{MM}(x,y)+I_{SM}(x,y)$$

The spatial intensity profiles of both beams were experimentally determined using appropriate beam diagnostic systems and expressed in absolute units (W/cm^2). Based on the superimposed distributions, the fraction of MM laser power within the SM beam diameter was obtained by spatial integration. The absolute laser powers were independently verified using calibrated power meters. It should also be noted that, although the influence of discrete SM laser beam power levels is applied in the work, the intensity of the SM laser beam changes slightly due to the constant focus position when it hits the front wall of the keyhole. To account for this effect, several SM power levels were investigated to compare the influence of increasing energy input and not specific power levels.

This analysis in Fig. 9 reveals that the additional MM laser beam energy present can be neglected with increasing SM power, since values below 2% are reached. Superpositions with additional lateral displacement ($\Delta_{S,y}$) achieve values that are correspondingly even lower. This enables a clearer interpretation of the later results. Differences between two process configurations can then only be attributed to the changed SM beam position, since the MM-laser contribution inside the SM laser beam focus range is negligible in both cases.

3. Results and discussion

The results of this study provide new and comprehensive insights into the fundamental influence of local laser beam intensity distributions on the keyhole shape and stability.

Particular attention is given to the role of the position of the

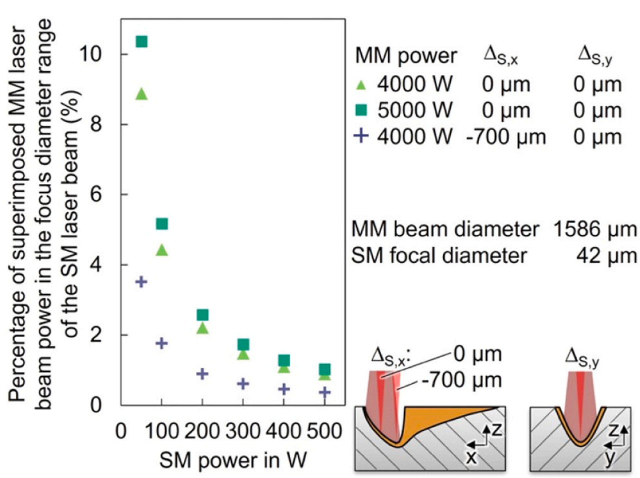


Fig. 9. Percentage of superimposed MM laser beam power in the focus diameter range of the SM laser beam reveals, especially at higher SM powers, that the increase in intensity due to superposition with the MM laser beam has no significant effect on the process and can be neglected.

additional intensity maximum within the keyhole, as significant differences were especially observed between the position on the front wall of the primary keyhole ($\Delta_{S,x}$ = 0 μm) and the position on the bottom ($\Delta_{S,x}$ = 700 μm).

The metallographic cross-sections in Fig. 10 show that in some cases a secondary seam cross-section area was formed (see Fig. 3 for explanation). Whether and to what extent depend on the SM power and the position of the SM laser beam. While these are only visible for the SM laser beam positions on the front wall of the keyhole from 300 W (4000 W MM laser beam power Fig. 10a and d) or 500 W respectively (5000 W MM laser beam power, Fig. 10c), they are already visible for the position on the primary keyhole bottom from 100 W (Fig. 10e and f). The samples welded with a lateral shift of the SM laser beam ($\Delta_{S,y}$) also show an eccentricity of the secondary seam cross-section peak whose values correspond approximately to $\Delta_{S,y}$ (see Fig. 11).

The mean primary seam cross-section area and the mean primary weld depth for the experiments in which the SM laser beam was set on the front wall of the primary keyhole increase significantly with increasing SM power (Fig. 12), while the values for the position on the keyhole bottom increase less in the case of the seam cross-section area and remain almost constant in the case of the mean primary weld depth. From this, it can be concluded that a certain proportion of the SM energy that hits the primary keyhole front wall is reflected there and then absorbed in the primary keyhole, causing the enlargement. However, if the keyhole bottom is hit directly, assumed at an average angle of approximately 90° and with a correspondingly high Fresnel absorption coefficient, a larger part of the SM energy can be absorbed there and used to generate the secondary seam cross-section area.

By subtracting the theoretical weld depth and seam cross-section area, obtained by linear superposition of the experimentally determined single-beam processes (see Fig. 4 for further explanation), from the corresponding values of the combined process, the validity of a simplified assumption that two keyhole shapes of different dimensions can be added together can be assessed. Based on the data shown in Fig. 13a, some values of the weld depth are comparable to those of the theoretical sum, and some are below. The deviation is particularly large for 5000 W MM laser beam power and the SM position on the front wall of the keyhole. The values with the SM position on the keyhole bottom ($\Delta_{s,x}$ = −700 μm, $\Delta_{s,x}$ = 0 μm and $\Delta_{s,x}$ = −250 μm) fluctuate about −250 μm below the value of the theoretical sum with large standard deviations.

The conditions at the bottom of the primary keyhole, such as the high temperature of the melt, should favor a high secondary weld depth. Nevertheless, the total weld depth is slightly below the theoretical sum of the addition of both individual processes. This systematic offset does not contradict the simplified superposition assumption but can be explained by several physical effects. First, attenuation of the SM laser beam by the vapor plume can reach several percent, depending on the process conditions [23], so that less SM power reaches the primary keyhole bottom. Second, partial interaction of the SM beam with the rear wall of the primary keyhole leads to additional intensity losses (see Fig. 17). Third, the dependence on process dynamics, which are evident here due to fluctuations and relatively high standard deviations. Fourth, the presence of a melt film at the bottom of the primary keyhole causes a systematic underestimation of secondary keyhole depth in metallographic cross-sections.

The thickness of this melt film can be estimated from the difference between the total weld depth determined metallographically and the mean keyhole depth measured by OCT for the single MM process without additional SM power. At an MM laser power of 4000 W, this difference amounts to approximately 240 μm (average weld depth: 2030 μm ± 49 μm; average keyhole depth: 1790 μm ± 150 μm, see also Supplementary Material 01). Consequently, secondary keyholes with depths below this value, such as those obtained at low SM power, cannot be resolved in metallographic sections, as they are fully embedded within the melt pool. This value would have to be added to all secondary

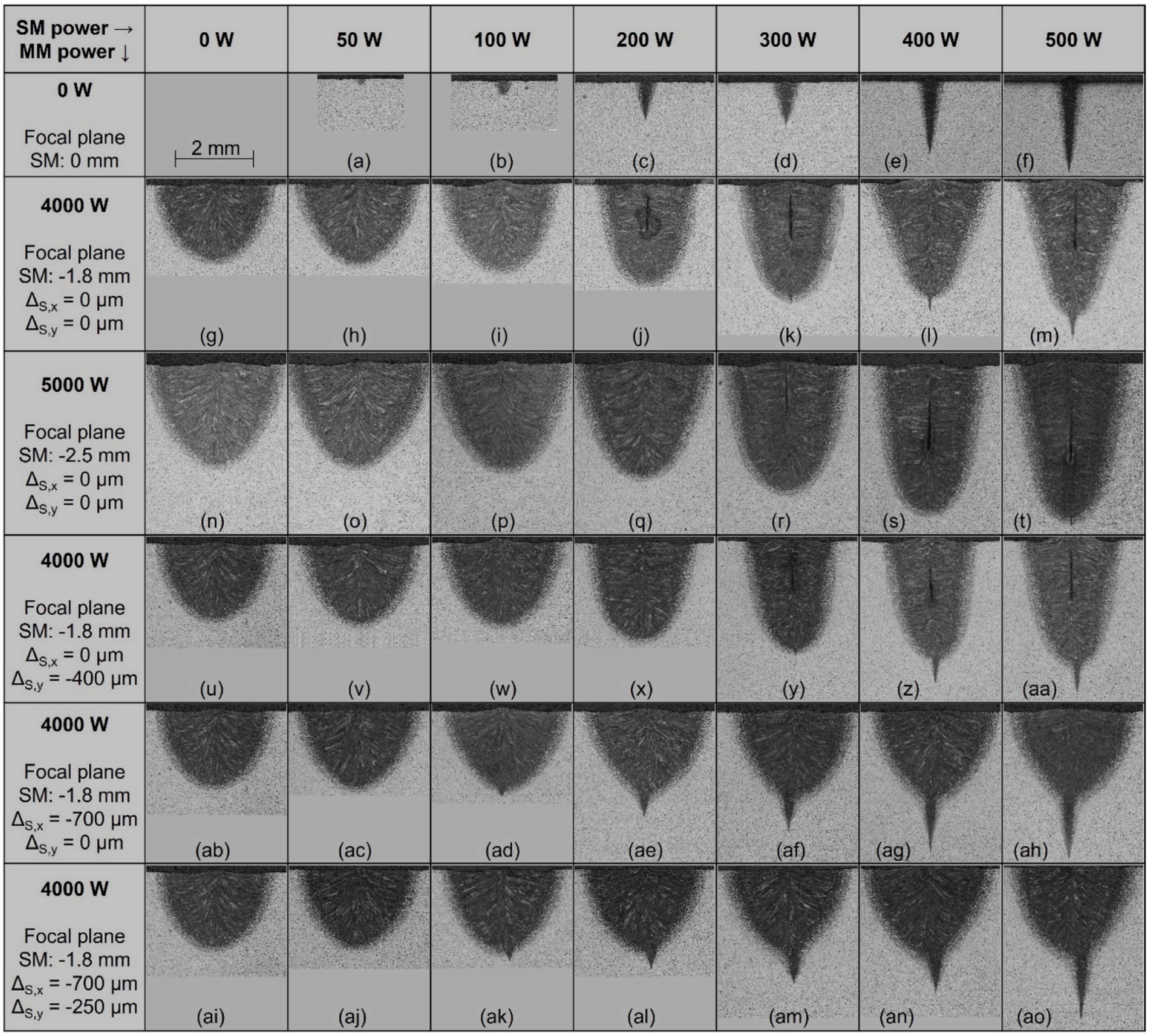


Fig. 10. Metallographic cross-sections.

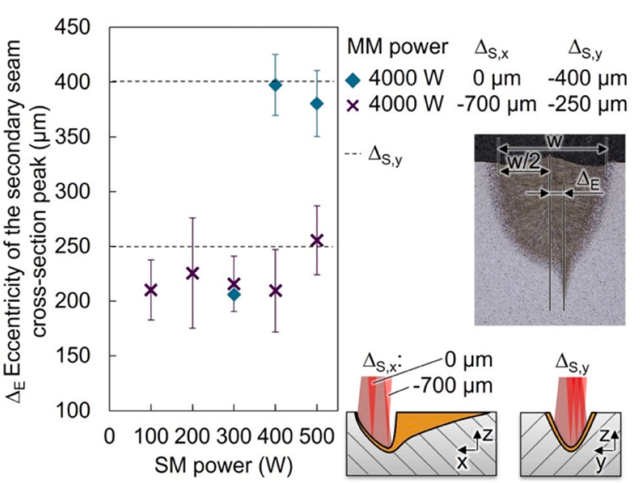


Fig. 11. Eccentricity of the secondary seam cross-section peak depending on SM power and position. Error bars represent ± SD from the mean value (n = 5 metallographic samples).

depths measured from the metallographic images to represent the actual secondary keyhole depth. It should be noted that the thickness of the melt film is subject to temporal fluctuations and reflects the uncertainty of the OCT and the cross-sectional depth measurements. It is not interpreted as a strictly constant physical thickness, but rather as a geometric correction term that accounts for the molten and subsequently solidified material beneath the primary keyhole bottom. When this correction is taken into account by adding 240 μm to the difference shown in Fig. 13, the corrected values closely approach a difference of 0 μm. Considering that additional attenuation mechanisms were not corrected for, this agreement supports and is thus very similar to those of the sum of the individual processes or even exceeds them in some cases.

Considering that additional attenuation mechanisms were not corrected for, the results support the assumption that two keyhole shapes of different dimensions can be added together in depth as simplification, especially if the secondary laser beam is set on the primary keyhole bottom. The values of the comparison of the seam cross-section areas, shown in Fig. 13b, indicate that the actual seam cross-section area is predominantly even larger than the theoretical sum of the values of the individual processes for all combinations examined. This suggests that the total absorption in the combined process is generally larger, and a higher efficiency can be reached. While the energy that escapes from the keyhole in the process with only the SM laser beam is lost to the surroundings (see Fig. 18b), it is, in the combined process, redirected into the primary keyhole, where it can be absorbed and contribute to the overall energy coupling (see Fig. 18f).

The analysis of the results obtained with OCT and high-speed imaging also shows that when the additional SM laser beam was positioned

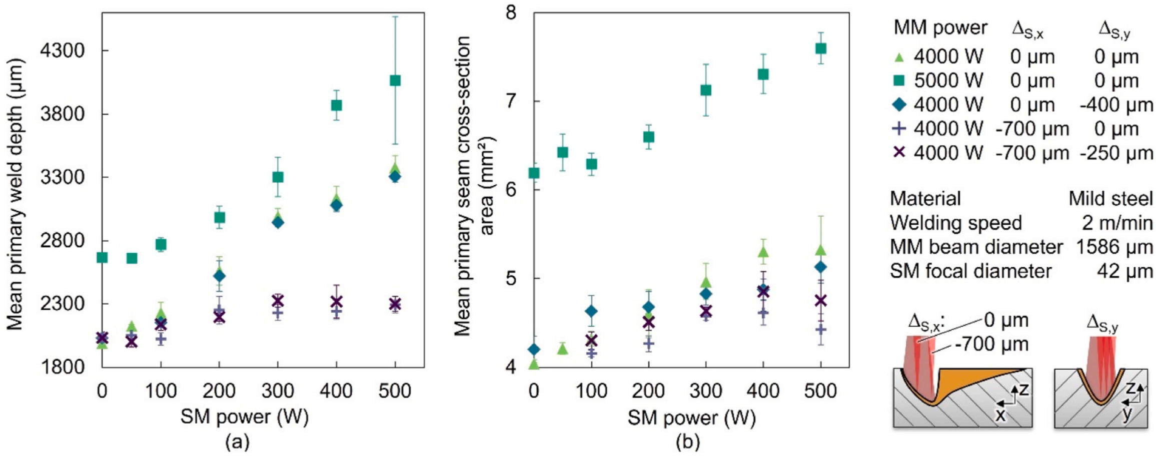


Fig. 12. Mean weld depth (a) and primary seam cross-section area (b) depending on the power of the SM laser beam and its position measured on metallographic cross-sections. Error bars represent ± SD from the mean value (n = 5 metallographic samples).

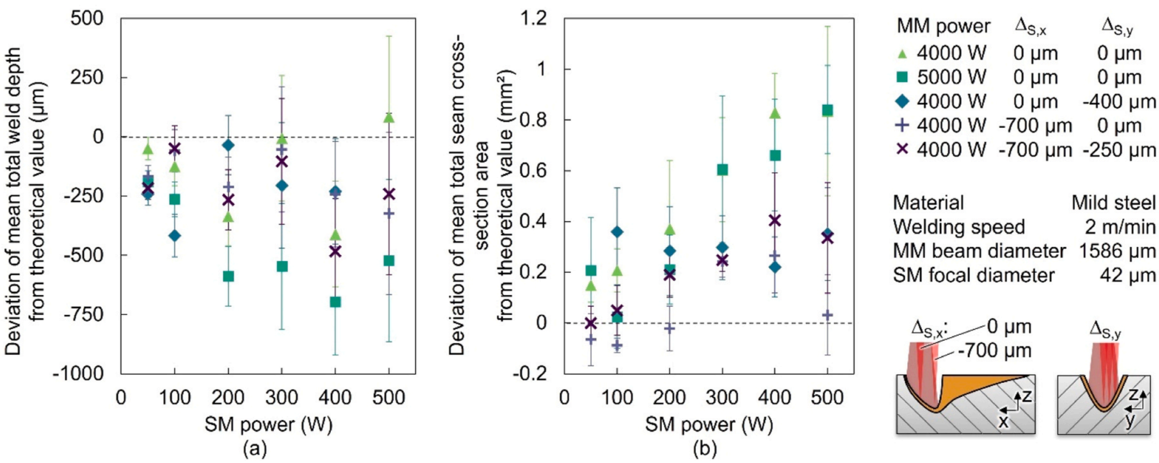


Fig. 13. Differences between values of the combined processes and the theoretical values. (a) Total mean weld depth and (b) total mean seam cross-section area, both subtracted by the mean theoretical value, which was determined by adding up the values of the individual processes (see Fig. 4 for further explanation). Error bars represent ± SD from the mean value (n = 5 metallographic samples).

on the front wall of the primary keyhole, strong coupling effects between the MM laser beam and SM processes were observed. The OCT measurements of the average distance of the primary keyhole bottom to the MM laser beam center (Fig. 14a) reveal that, with increasing SM power, the keyhole bottom shifted in the direction of the position of the SM laser beam. Up to this point, no significant difference could be identified

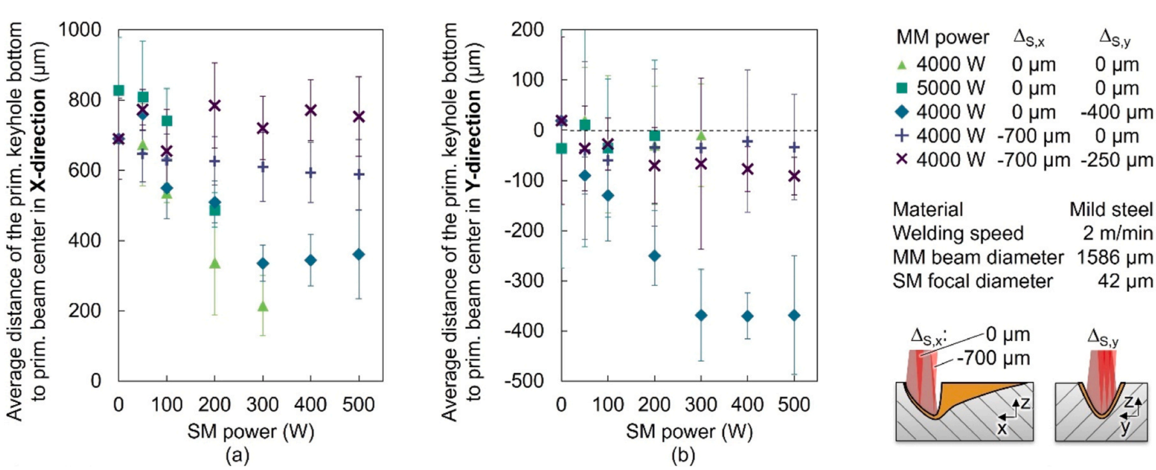


Fig. 14. Average distance of the primary keyhole bottom to the primary laser beam center in X- (a) and Y-direction (b) depending on the power of the SM laser beam and its position determined using OCT measurements. Error bars represent ± SD from the mean value (n = 5 OCT-measurements).

between the previous results with and without lateral deviation ($\Delta_{S,y}$) of the SM position, except for the eccentricity of the secondary suture cross-section peak. However, the analysis of the average distance of the primary keyhole bottom to the primary beam center in the Y-direction shows a corresponding additional displacement of the keyhole bottom in the Y-direction. This can be explained by the fact that the additional intensity was absorbed predominantly laterally shifted at the front wall of the primary keyhole, thus forming a preferred keyhole expansion. Although the primary keyhole bottom at 5000 W MM laser beam power is further away from the MM laser beam center (830 μm) than that at 4000 W MM laser beam power (690 μm), both approach the center similarly as the SM power increases.

Since the high-speed videos show that the front of the keyhole opening remains in the same position even as the SM power increases (see Fig. 15c and d for example), whereas the primary keyhole bottom shifts forward, it can be concluded that the additional intensity on the front wall of the keyhole reduces its inclination angle, resulting in a steeper front wall. This change in geometry affects the internal redistribution of the laser radiation within the keyhole. According to the law of reflection, less of the laser beam radiation reflected at the front wall of the keyhole is to be reflected toward the rear wall [24]. In addition the angle of the vapor plume is also reduced and less directed toward the rear wall [25]. As a result, the length of the keyhole opening is reduced, which is confirmed for both used MM laser beam powers by the high-speed camera-based measurements (see Fig. 15a). For 4000 W MM laser beam power, the mean length of the keyhole opening was reduced from 1.4 mm to 1.1 mm with 500 W MM laser beam power, for 5000 W MM laser beam power it was reduced from 1.6 mm to 1.3 mm. Both correspond to a similar reduction of approximately 20%. Since the widths of the keyholes remain constant, the reduction in length results in a decrease in the total size of the keyhole opening area (see Fig. 15b) in both cases.

As mentioned, measurements of keyhole shape using OCT and cross-section images only show the average geometric ratios. When looking at the cross-sections and the average keyhole shapes that are obtained with the SM position at the keyhole front wall, the question arises if the shifted keyhole constantly retains its shape or whether it fluctuates.

With increasing SM power and increasing shift of the keyhole bottom toward the front, compared to the results with no additional SM power (see Fig. 8) all histograms of the OCT measurements show increasing secondary maxima (see Fig. 16). These secondary maxima are located slightly above the depth range corresponding to the original position of the primary keyhole bottom (see also Supplementary Material 01). This indicates that the keyhole is either formed in a shape with a shallower and a deeper keyhole bottom (and the additional secondary keyhole) or oscillates between these two states.

The high-speed recordings suggest a very dynamic rather than a static process behavior. The point of impact of the SM laser beam oscillates up and down the keyhole front wall (see Fig. 17d, e, and f for example), presumably depending on the current inclination angle and the resulting position of impact.

It is therefore assumed that the processes with the SM position on the primary keyhole front wall fluctuate between the two states shown in Fig. 18, where the SM laser beam is either absorbed at the forward shifted primary keyhole bottom (Fig. 18c) or forms a secondary keyhole on the front wall (Fig. 18d). This is also supported by OCT data combined with the metallographic cross-sections. Since at a MM laser beam power of 4000 W, 300 W SM power is already sufficient to shift the primary keyhole bottom near the position of the SM laser beam (see Fig. 14), the excess energy should actually form a secondary keyhole at the primary keyhole bottom, which should be visible in the cross-sections. In fact, however, a small secondary seam cross-section area is only visible for 500 W SM power (Fig. 10c). This suggests that although SM power may have created a secondary keyhole, it is probably located partly on the front wall of the keyhole and does not extend beyond the depth of the primary keyhole and is therefore not visible in the cross-section. Since the beam diameter of the OCT is bigger than the secondary keyhole opening diameter, only the shape of the primary keyhole can be measured using OCT.

While the process behavior of the individual processes is rather stable and only a few spatters are emitted, the significant increase in the average number of spatter and the resulting overall spatter amount with increasing SM power on the keyhole front wall and the increasing MM laser beam power from 4000 W to 5000 W also indicate a lower stability or a higher process dynamic (see Fig. 19) respectively. Based on additional high-speed recordings and adjusted illumination settings, as illustrated by the representative image sequence in Fig. 19d, e and f, it was observed that, for parameter combinations in which the SM laser beam was positioned on the front wall of the primary keyhole, spatter formation predominantly originated from the lateral regions of the keyhole. This observation indicates that the spatter was mainly generated by recoil pressure– or melt flow–driven mechanisms rather than by

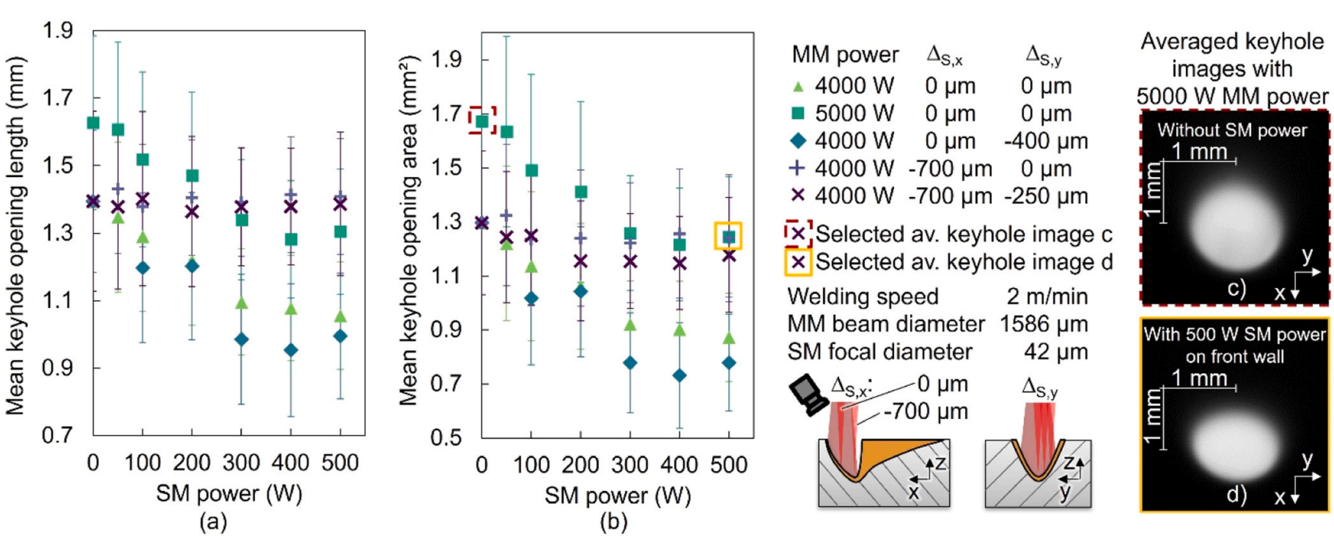


Fig. 15. Mean keyhole opening length (a) and mean keyhole opening area (b) depending on the power of the SM laser beam and its position, c) averaged keyhole image of the process with 5000 W MM laser beam power with no additional SM power, and d) the 500 W SM laser beam set at the keyhole front wall showing the shortening of the keyhole opening. Error bars represent ± SD from the mean value (n = 20,000 images from one video file). The video files are available at supplementary materials Figure_02 and 03].

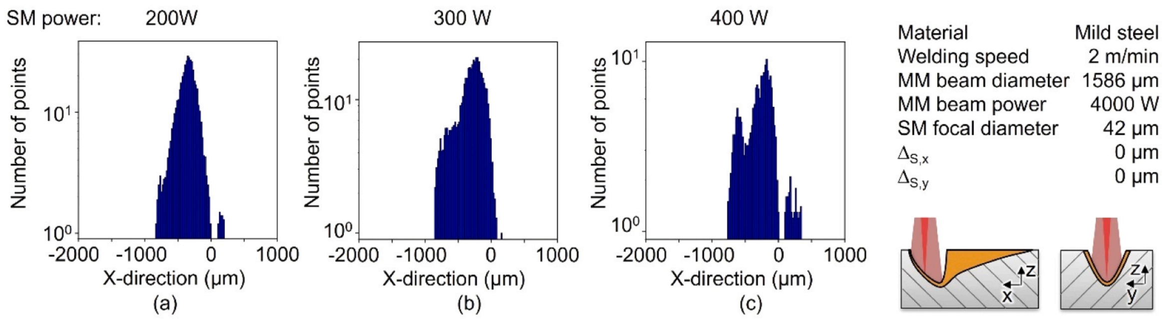


Fig. 16. Example of increasing secondary maxima with increasing SM power set on the front wall of the primary keyhole. Histograms of the averaged keyhole bottom positions (n = 5 OCT-measurements), see also Fig. 8f. It should be noted that some OCT measurement points with 400 W SM power are below the 3000 µm range and are therefore not visible in the histogram. Nevertheless, the image illustrates the formation of the secondary maximum, which is located at a depth above 3000 µm and is therefore valid (see also Supplementary Material 01).

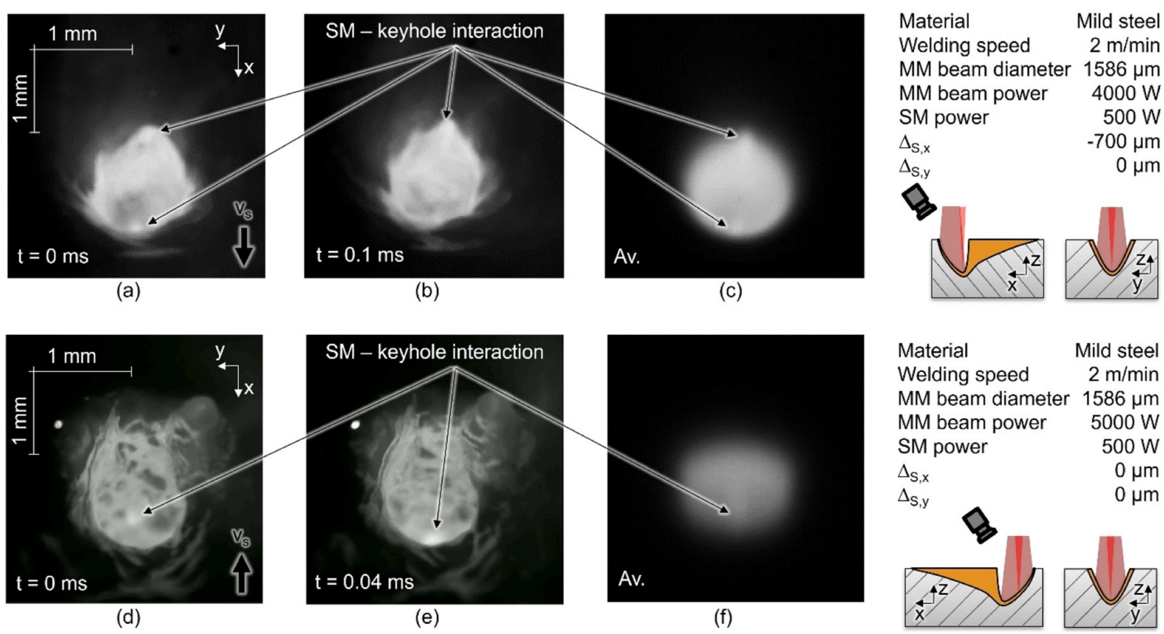


Fig. 17. Interaction of the SM laser beam with the primary keyhole. (a, b) High-speed video images and (c) averaged (Av.) image of the SM interaction at the bottom and the rear wall of the keyhole. (d, e) High-speed video images and (f) averaged (Av.) image of the SM interaction at the front wall of the keyhole. The video files are available at supplementary material Figure 03 and 04.

a collapse of the keyhole. The increased spatter amount can presumably be attributed to the oscillation of the process between the different aforementioned states. It is also suspected that the SM radiation reflected from the front wall toward the rear wall of the primary keyhole causes the rear wall to bulge, which also causes an unstable process and can also lead to spattering [26]. The additional reduction in the size of the keyhole opening results in restricted vapor flow and also encourages spattering [15]. The average single spatter volume (Fig. 19b) is not significantly affected by the changed parameters.

The results with the SM position at the front wall of the keyhole partly reject the hypothesis that the properties of individual laser beam processes can exist independently within a combined process. In this context, the present findings can be directly related to the phenomena reported in [10], which likewise employed a coaxial superposition of a SM and a MM laser beam. In that earlier study, we hypothesized that the secondary SM beam is largely transmitted through the primary keyhole and acts predominantly at the keyhole bottom, based on a concentric beam position ($\Delta_{S,x} = \Delta_{S,y} = 0$ µm) and the assumption of a rotationally symmetric keyhole. However, due to the absence of in-situ measurements of the keyhole shape at that time, the actual interaction location could not be verified. The results of the present study show that, under comparable conditions, this beam position most likely corresponds to an interaction with the front wall of the primary keyhole rather than with its bottom. This reinterpretation is consistent with the significant increase in primary weld depth observed both in [10] and in the present work with increasing SM intensity applied at the front wall. Moreover, similar effects were reported for aluminum in [10], indicating that the underlying interaction mechanism is not restricted to S235JRC steel. However, the quantitative expression of the observed effects is expected to depend on material-specific thermophysical properties such as absorption behavior and vaporization characteristics. Therefore, the transferability of the results should be understood in a qualitative sense, while further investigations are required for other material classes. The additional SM intensity has a fundamentally different effect when placed on the primary keyhole bottom at $\Delta_{S,x}$ = -700 µm. The secondary SM laser beam formed a secondary keyhole volume at the bottom of the primary keyhole without noticeably altering the overall primary keyhole shape. This is supported by quantitative measurements showing that the primary keyhole depth (Fig. 10b), the position of the keyhole bottom (Fig. 13), and the size of the keyhole opening (Fig. 15) remain

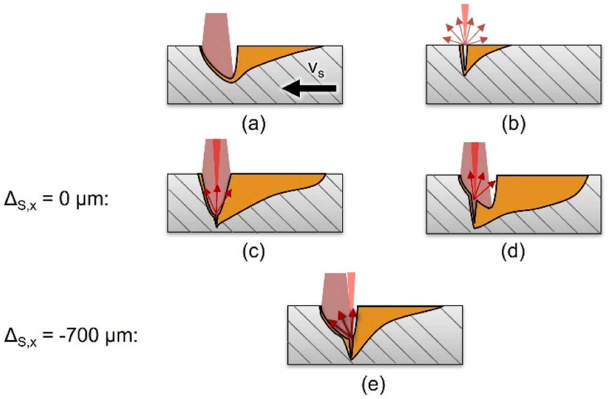


Fig. 18. Schematic longitudinal sections of the welding processes. (a) Resulting keyhole with only the MM laser beam. (b) Process with only SM laser beam and partially lost energy. (c) Process with additional SM laser beam at $\Delta_{S,x} = 0$ µm resulting in a secondary keyhole located at the forward-shifted bottom of the primary keyhole. (d) Process with additional SM laser beam at $\Delta_{S,x} = 0$ µm resulting in a secondary keyhole located at the front wall of the primary keyhole. e) Process with additional SM laser beam at $\Delta_{S,x}$ = -700 µm resulting in a secondary keyhole located at the bottom of the primary keyhole.

nearly constant as the SM laser power increases. Because both the position of the front keyhole opening and the position of the keyhole bottom remain unchanged, it can also be concluded that the inclination angle of the front wall of the primary keyhole is not affected. If the thickness of the bottom melt film is taken into account, the weld depths and seam cross-section areas resemble the values of the theoretical sum of the individual processes or even exceed them. The process stability also remained unchanged, as shown by the spatter evaluations (Fig. 19).

When the SM laser beam is set to the primary keyhole bottom, almost no spatter can be seen even at 500 W SM power. This can also be explained by the fact that the primary keyhole remains largely unaffected, which confirms in part the hypothesis that the characteristics of individual laser beam processes can coexist independently within a combined process.

Although the additional shift of the SM position in the Y-direction ($\Delta_{S,y}$) is visible in the metallographic cross-sections (see Fig. 10f), all other measured values show very similar results to the values of the process without the additional Y-shift mentioned above. This can be explained by the fact that with a shift of $\Delta_{S,y}$ = -250 µm the SM laser beam still predominantly hits the outer edges of the bottom of the primary, as the inclination angle of the lateral keyhole walls is much steeper than those of the keyhole front wall. The secondary keyhole can be formed in this position, which is supported by the eccentricity that corresponds to approximately 250 µm (see Fig. 11). These findings also support the hypothesis by showing that the position of the secondary keyhole does not depend on the central intensity maximum of the primary laser beam or on multiple reflections within the primary keyhole.

The results of this study demonstrate that detailed knowledge of the keyhole shape is essential for understanding the resulting process behavior. In particular, the exact region within the keyhole where additional intensity maxima interact plays a decisive role. This region can be identified using OCT measurements and high-speed imaging. The systematic variation of the relative position of a localized intensity maximum within the keyhole shows that this position constitutes a key control parameter and can be more influential than the absolute intensity itself. Based on the presented findings, general guidelines for beam optimization can be derived. An intensity maximum acting at the primary keyhole bottom allows a largely independent superposition of individual laser beam processes and results in a stable process behavior. An interaction at the front wall of the keyhole leads to pronounced

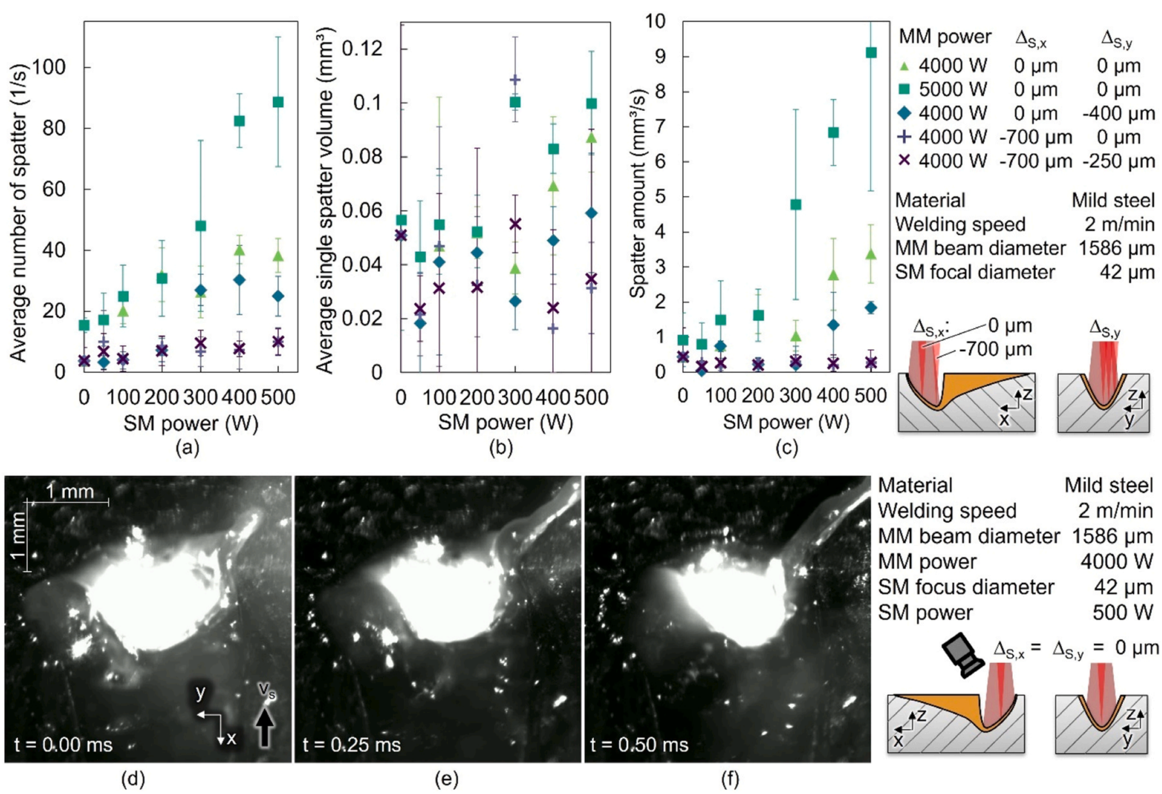


Fig. 19. Evaluation of spatter behavior. (a) Average number of spatter. (b) Average single spatter volume. (c) Average spatter amount as the result of the multiplication of (a) and (b). Error bars represent ± SD from the mean value (n = 80,000 images from four video files). The representative high-speed video image of (d) – (f) shows that the spatter predominantly detaches from the lateral areas of the keyhole. The corresponding video file is available at supplementary materials Figure 06. Representative spatter tracking video files are available at supplementary materials 07 to 13.

changes in keyhole shape and process behavior. Therefore, if a strong modification of the primary process is desired, the additional intensity should be positioned at the keyhole front wall. In contrast, if the goal is to maintain process behavior and for example increase penetration depth, the intensity maximum should be positioned at the keyhole bottom. These insights do not aim at optimizing a specific welding application but rather provide a fundamental understanding of how localized intensity features couple with the keyhole. The results establish a basis for the future understanding-based design of task-specific laser beam intensity distributions, enabling targeted manipulation of keyhole behavior.

4. Conclusions

This study conducted fundamental investigations regarding the influence of laser beam intensity maxima and their positions on deep penetration laser welding induced keyholes. By using an optical system that coaxially combines a MM laser beam and a SM laser beam, the MM laser beam was used to create a large primary keyhole in which the beam of the SM laser source formed an intensity maximum and a secondary keyhole that was shifted throughout the experiments. The corresponding effects on the primary keyhole shape and its stability were investigated. The following conclusions contribute to the possibility of a comprehension-based, targeted local manipulation of the keyhole:

1. Knowledge of the shape of the keyhole, the precise position of the keyhole bottom and the area (front wall, bottom, etc.) in which additional intensity maxima interact is crucial for a targeted manipulation of the keyhole and the investigation of observed phenomena. It has been demonstrated that OCT measurements and high-speed imaging are suitable for determining these regions.
2. The hypothesis that a second, significantly smaller keyhole volume can be produced and positioned largely independently of the main primary keyhole was confirmed for the combination of a primary laser beam diameter of 1586 μm and a secondary laser beam diameter of 42 μm.
   a. Provided that the additional secondary intensity is directed at the area of the primary keyhole bottom, the coaxial secondary laser beam generates a secondary keyhole at the bottom of the primary keyhole, which itself remains largely unaffected. The simplified assumption that the two known keyholes from the respective individual processes can be "added" was validated. This indicates that the approach can be suitable for the design of the intensity distribution under the given conditions. This also applies to intensity maxima that are not located on the axis of the welding direction but slightly lateral off-center on the primary keyhole bottom, as it does not depend on the central intensity maximum of the primary laser beam or on multiple reflections within the primary keyhole.
   b. However, an additional single intensity maximum on the front wall of the keyhole significantly influences the primary keyhole. As a result, the inclination angle of the front wall of the primary keyhole changed in such a way that the keyhole bottom shifts toward the added intensity maximum in welding direction as the intensity increases, also changing the length of the keyhole opening. This produces an oscillating rather than a stable keyhole shape, which means that increasing intensities are accompanied by increased process irregularities such as spatter. The findings also apply to intensity maxima that are not located on the axis of the welding direction but slightly lateral off-center on the keyhole front wall, which can cause a shift of the keyhole bottom forward and sideways.

CRediT authorship contribution statement

Peer Woizeschke: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Thomas Seefeld:** Writing – review & editing, Resources. **Ronald Pordzik:** Writing – review & editing, Software. **Marcel Möbus:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jmatprotec.2026.119323.

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